

DC: Research on Augmented Reality

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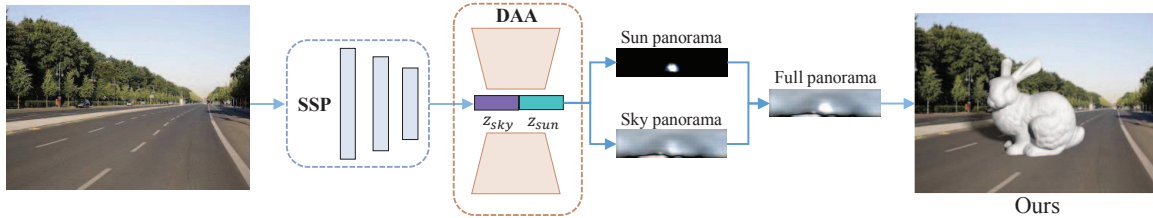


Figure 1: The data-driven sun and sky model can preserve both the high dynamic intensity of the sun and the complex environment.

ABSTRACT

Augmented reality technology can integrate virtual information (objects, pictures, videos, sounds, etc.) into the real environment, enrich the real world, and build a more comprehensive and better world. Outdoor illumination is essential for AR applications. We will discuss our proposed two methods which predict outdoor illumination from a single outdoor image. With predicted illumination, the virtual object can be inserted into the real environment, with consistent shadow information and occlusion relationship.

Index Terms: Computing methodologies—Computing methodologies—Computer vision; Human-centered computing—Visualization—Visualization application domains

1 OUTDOOR ILLUMINATION

Outdoor illumination prediction based on a single input image is a key task for augmented reality (AR). However, the diverse weather conditions and the complex interaction between illumination and other scene properties (*e.g.*, surface reflectance and geometric variations) make this problem highly challenging. We have proposed two learning-based methods to predict outdoor illumination and achieve state-of-the-art performance by employing powerful neural networks.

1.1 Dual attention autoencoder

Accurate outdoor illumination estimation is not easy due to the extremely complicated sky appearance and the mutual interference of the sun and sky. In view of these challenges, we present a deep approach [2] for the estimation of all-weather outdoor illumination. The key to the approach is a novel dual attention autoencoder (DAA) with two independent branches to compress the sun and sky lighting information from an input HDR panorama, respectively. This enables more accurate lighting estimation as evidenced by experiments since the mutual interference between the sun and sky can be precluded effectively. In DAA, we design the adaptive feature pyramid and the attention module to promote its accuracy in compression. We further develop a sun-sky predictor, a masked network, to learn the sun and sky lighting conditions from an FoV-limited image. The full pipeline is shown in Fig. 1.

The method generates coherent outdoor lighting and provides plausible shadings and shadows as the viewpoint changes, enabling

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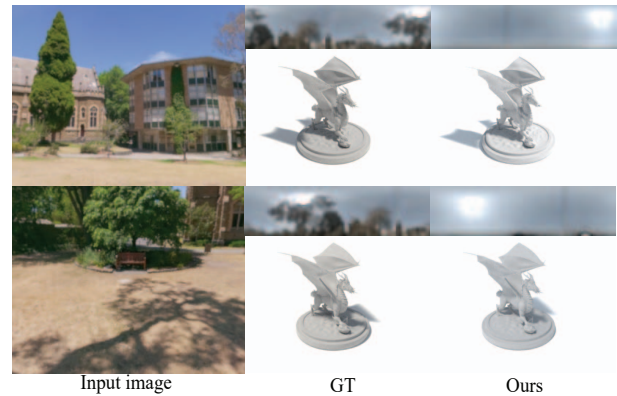


Figure 2: Visual results of our SSP for different viewpoints from the same panorama.



Figure 3: Demonstration of virtual object insertion. Different virtual objects (the bench, horse and dragon) are inserted into real scenes with consistent shading and shadowing.

smooth transitions from one viewpoint to another. In Fig. 2, we predict outdoor illumination for 4 FoV-limited images from the same panorama.

Fig. 3 shows some examples of virtual object insertion. These scenes further prove that our method can accurately predict the illumination of many outdoor images under different weather conditions.

1.2 HDSky

Our first method only disentangles outdoor lighting into the sun and sky. To obtain the fine-grained control over independent lighting factors of the HDR intensity of the sun and the complex textures of the surrounding sky, we propose a hierarchical disentangled sky model (HDSky) [1] for outdoor illumination prediction. The full pipeline is shown in Fig. 5. With this model, any outdoor panorama can be hierarchically disentangled into several factors based on three well-designed autoencoders. The first autoencoder compresses each sunny panorama into a sky vector and a sun vector with some con-

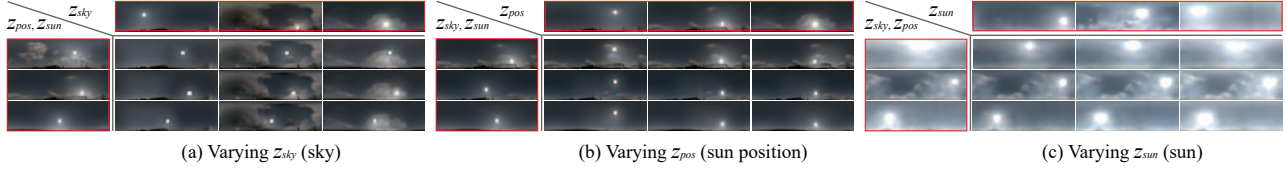


Figure 4: Varying a single lighting factor.

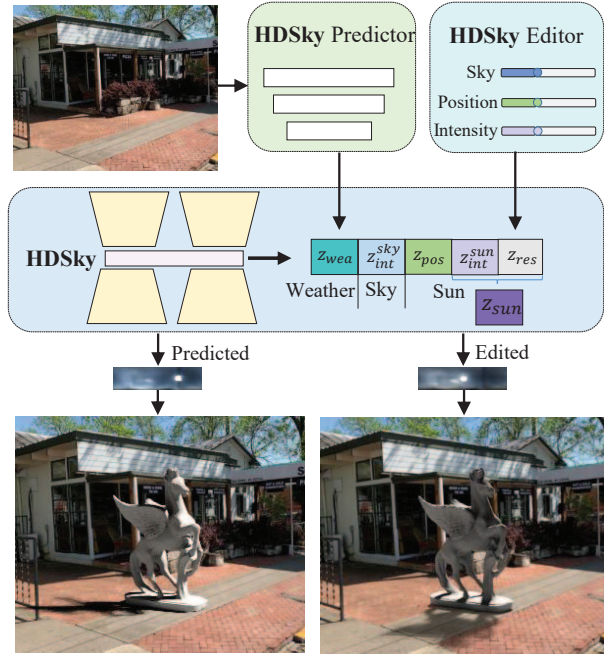


Figure 5: HDSky disentangles the learned latent vector into several independent and meaningful factors, enabling intuitive illumination editing with physical plausibility.

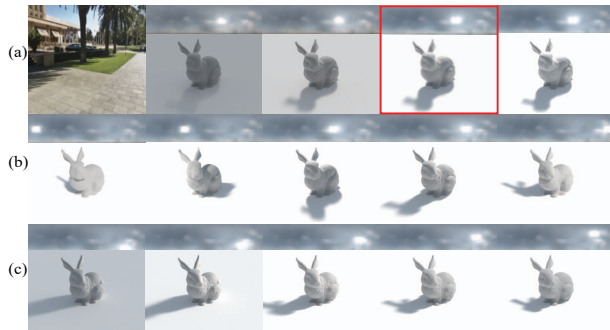


Figure 6: Examples of intuitive edits of our HDSky editor. We edit (a) the sun intensity, (b) sun azimuth and (c) sun elevation, respectively.

straints. The second autoencoder and the third autoencoder further disentangle the sun intensity and the sky intensity from the sun vector and the sky vector with several customized loss functions respectively. Moreover, a unified framework is designed to predict all-weather sky information from a single outdoor image. Through extensive experiments, we demonstrate that the proposed model significantly improves the accuracy of outdoor illumination prediction.

We further evaluate how well HDSky disentangles each factor (the sky vector z_{sky} , the sun position z_{pos} and the sun vector z_{sun}) and generates realistic panoramas. Our disentanglement of each



Figure 7: Demonstration of virtual object insertion. Benefited from our HDSky, different virtual objects (the horse and dragon) are inserted into real scenes with consistent shading and shadows.



Figure 8: Virtual object insertion.

factor on HDR panoramas is shown in Fig. 4. For each subfigure, the panoramas in the top row and leftmost column (with red boxes) are reconstructed panoramas. The specific factors taken from each reconstructed panorama are indicated in the top-left corner.

Our HDSky editor provides explicit parameters for users to edit the predicted panorama. The visually smooth transitions are shown in Fig. 6. Fig. 7 shows that the estimated lighting intensity of our HDSky predictor can provide plausible shadows on real-world images under different weather conditions.

2 VIRTUAL OBJECT INSERTION

Outdoor illumination prediction based on a single input image is a key task for many applications, such as AR. For virtual objects inserted into real images, illumination is necessary to generate the bidirectional shadows which contain real-to-virtual and virtual-to-real shadow rendering. Similar to the bidirectional shadows, the positional relationship between virtual and real objects is also bidirectional (see Fig. 8). AR should transformatively blur the line between reality and the virtual world. In the future, we will learn to insert virtual objects with occlusion-aware AR effects and shadows which are consistent with the whole real scene.

ACKNOWLEDGMENTS

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